Computer

↳ A computer is an electronic device that accepts data (input), processes it according to a set of instruction (program), stores it for future use, and produces meaningful information (output).

↳ A computer is a programmable electronic device that executes instructions stored in memory.

↳ It receives data via input devices, processes it using a CPU, stores results in memory or storage units, and displays output through output devices.

↳ It uses binary language (0s and 1s) internally.

☆ Electric Devices

- Definition: Devices that convert electrical energy into other forms of energy like heat, light or motion.
- Working principle: Operate mainly on the flow of electric current.
- Components: Use basic components like wires, switches, motors, and resistors.

- Control : Usually not intelligent - no processing or automatic control.

- Examples : Electric fan, bulb, heater, iron, refrigerator.

☆ Electronic Device

- Definition : Devices that control the flow of electrons using semiconductors to perform specific tasks like computing, signal processing, or communication.

- Working principle : Operate using electric signals and data manipulation.

- Components : Use complex components like transistors, diodes, ICs, microprocessors.

- Control : smart and capable of processing, memory, and automation.

- Examples : Computer, mobile phone, television, calculator.

↳ A semiconductor is a material that has electrical conductivity between a conductor (like copper) and an insulator (like glass).

↳ Semiconductor can act as both a conductor and an insulator, depending on conditions such as temperature, voltage, or the presence of impurities.

• **General Functions of a computer**

① Input

↳ Receives raw data and instructions from input devices (e.g ÷ keyboard, mouse, scanner)

② Processing

↳ The CPU processes the input data according to the program or instructions.

③ storage

↳ Temporarily holds data (RAM) or permanently stores data and program (hard drive, SSD).

④ Output

↳ Delivers the processed information through output devices (e.g. monitor, printer, speaker).

⑤ Control

↳ Directs the manner and sequence in which all operations are carried out using the control unit.

• Main components of a Computer

① Input Devices

↳ Used to enter data and instructions into the computer.

↳ e.g: keyboard, mouse, scanner, microphone, webcam

② Output devices

↳ Display or present the result of computer processing.

↳ e.g: Monitor, Printer, speakers, Projector.

③ Central Processing Unit (CPU)

↳ The "brain" of the computer that processes all data and instructions.

- Arithmetic Logic Unit (ALU): Performs calculations and logic operations.

- Control unit (CU): Directs operations of the processor.

↳ Processor types: Intel, AMD, Apple M-series

④ Memory / storage

↳ Primary Memory (RAM): Temporary memory used during processing.

↳ Secondary storage: Permanent data storage.
e.g: Hard Disk Drive (HDD), SSD

⑤ Motherboard

↳ Main circuit board that connects all components.

↳ Contains CPU socket, RAM slots, expansion cards, and power connectors.

⑥ Power Supply Unit (PSU)

↳ Converts electricity from wall outlets into usable power for the computer.

⑦ Cooling system

↳ Keeps components like CPU and GPU from overheating.

↳ e.g ÷ Fans, heat sinks, liquid cooling

⑧ Software (System & Application)

↳ System software: Operating system (windows, macOS, Linux).

↳ Application Software: Programs like Ms word, browsers, games.

⑨ Expansion Cards (optional).

↳ Enhance or add features to a computer.

↳ Examples: Graphics card (GPU), Sound card, network card.

• Characteristics of a computer

① Speed

↳ A computer can process millions of instructions per second.

↳ Tasks that take humans hours can be done in seconds.

↳ Measured in MHz (megahertz), GHz (gigahertz), or FLOPS (floating-point operating per second).

Speed / Time Units Table

Unit	Symbol	Power of 10
Milisecond	ms	10^{-3} seconds
Microsecond	μs	10^{-6} seconds
Nanosecond	ns	10^{-9} seconds
Picosecond	Ps	10^{-12} seconds
Femtosecond	fs	10^{-15} seconds

② Accuracy

↳ Computers perform tasks with high precision and very few errors.

↳ Errors usually occur due to incorrect input or programming, not the machine itself.

③ Automation

↳ Once programmed, a computer can perform tasks automatically without human intervention.

④ Storage

↳ Computers can store vast amounts of data and instructions.

↳ Data can be retrieved instantly when needed.

↳ Includes both temporary storage (RAM) and permanent storage (HDD/SSD)

- Table of storage measurement units.

Unit	Symbol	Value
Bit	b	1 bit
Nibble	-	4 bits
Byte	B	8 bits
kilobyte	KB	1024 Bytes
Megabyte	MB	1024 KB
Gigabyte	GB	1024 MB
Terabyte	TB	1024 GB
Petabyte	PB	1024 TB
Exabyte	EB	1024 PB
Zettabyte	ZB	1024 EB
Yottabyte	YB	1024 ZB

5. Versatility
↳ Can perform a wide range of tasks: typing, calculating, designing, gaming, Coding, etc.

↳ A single computer can switch between different applications quickly.

6. Multitasking
↳ Modern computers can run multiple applications at once.

↳ For example, browsing the internet while playing ~~musinr~~ music and downloading files simultaneously.

7. Diligence
↳ Unlike humans, computers do not get tired, bored, or distracted.

↳ They can work continuously for hours or days without loss of performance.

8. Communication
↳ Computers can easily communicate with other devices or networks.

↳ Used in emailing, chatting, video calls, and internet browsing.

9) No intelligence

→ Computer cannot think or make decisions on its own.

→ It requires specific instruction (programs) from users to perform any task.

10) Programmability

→ A computer can be programmed using software to perform a wide range of tasks.

→ This makes it flexible and reusable for many purposes.

- Example Interpretation
  - A CPU with 3.2 GHz means it performs 3.2 billion cycles per second.
  - An SSD with 500 MB/s speed means it can read or write 500 megabytes every second.

- <u>Advantages of computer</u>

> High Speed - Computers can process data and perform tasks in milliseconds or less.

> Accuracy - They perform calculations and tasks with near-perfect precision, reducing human error.

- Storage - Vast amount of data can be stored and retrieved quickly.
- Automation - Repetitive tasks can be automated, saving time and effort.
- Multitasking: Computers can run multiple application simultaneously.
- Connectivity: They enable fast communication via the internet (emails, video calls, etc)
- Data Analysis - Computers can analyze and visualize large datasets efficiently.
- Cost-Effective - In the long run, they reduce labor and operational cost.
- Research and learning - Access to online resources and educational tools enhances knowledge.
- Entertainment - Used for games, movies, music, and social media.

Disadvantages of Computer

> Health Issues - Prolonged use can cause eye strain, poor posture, and repetitive strain injuries.

> Cyber Security Risks - Computers are vulnerable to viruses, hacking, and data breaches.

> Job displacement - Automation can lead to unemployment in certain sectors.

> High dependency - Over-reliance on computers may reduce problem solving or thinking skills.

> Privacy Concerns - Personal data can be misused or tracked without consent.

> Social Isolation - Excessive computer ~~can~~ use can reduce face to face interaction.

> Environmental Impact - E-waste and energy consumption harm the environment.

> Initial cost - High-performance computers and maintenance can be expensive.

> Distraction - Games, social media, and Entertainment can reduce productivity.

> Obsolescence - Technology becomes outdated quickly, requiring frequent upgrades.

- 10 Applications of computer in ~~Relati~~ Relevant Fields

1) Education
- Computer are used in the education sector for teaching, learning, and administrative purpose. This provid access to online courses, e-books, educational videos and virtual classrooms, etc.

2) Healthcare:
- In hospitals and clinics, computers are used to maintain electronic medical records, schedule appointments, assist in diagnosis through imaging technologies like CT scan and MRIs, and support telemedicine for remote consultations.

3) Business and Commerce
- Businesses use computers for managing accounts, costumer data, inventory, marketing, and communication. E-commerce platforms, online payment systems, and business analytics heavily rely on computer technology.

4) Science and Research
- Computer play a crucial role in scientific research by processing complex data, running simulations, analyzing results, and enabling discoveries in fields such as physics, chemistry and space science.

5) Government and Administration
- Government offices uses computers to store and manage citizen data, process official documents, handle tax records, and offer public service online.

6) Banking and Finance
- Computers are essential in the banking sector for handling transactions, maintaining customer accounts, managing ATMs, conducting online banking, and analyzing financial trends and risks.

7) Communication
- Computers enable instant communication through emails, messaging platforms, and

video conferencing. Computer helps to connect people around the world.

8) Engineering and Design

- Engineers and designers use computer software for 2D and 3D modeling, simulations, architectural planning, and structural analysis, making the design process fast, more accurate and efficient.

9) Entertainment

- The Entertainment industry uses computers to produce movies, music, video games, and animations.

10) Transportation

- In transportation, computers are used in vehicle navigation systems, airline and railway reservations, traffic control, and the development of self-driving vehicles, contributing to safer and more efficient travel.

Input, Process and Output

Input → Process → Output
↑↓
Storage

1. Input Unit

- The input unit is responsible for feeding data and instructions into the computer system for processing. It translate human-understanding information (like text, sound, images) into a machine-readable format. (binary)

- Functions

> Accepts data & instructions from the user or external sources.

> Convert data into binary format (0s and 1s) understandable by the CPU.

> Transfers data to the main memory for processing.

- Examples of Input Devices

1. Keyboard - For entering text, numbers and commands.

2. Mouse - For pointing, clicking, and selecting

3. Scanner - For converting printed documents /images into digital form.

4. Microphone - For recording audio input.

2. <u>Processing Unit (CPU)</u>

- The Central Processing Unit (CPU) is the brain of the computer. It processes all data and instructions received from the input unit and produces the required output.

- <u>Main Components of CPU</u>

a) Control Unit (CU)

↳ The control unit is the nerve center of the CPU that manages and directs how the computer's various components work together.

↳ It does not process data itself, but it controls the flow of data between the CPU, memory, and input / output devices.

↳ It acts like a traffic controller.

↳ Decides what to do (based on the instructions fetched from memory).

• <u>Main Functions</u>

1. Instruction Fetching

- Retrieves (fetches) program instruction ~~into signals that oth~~ from main memory

(RAM).

2. Instruction Decoding
- Interprets (decodes) the fetched instruction into signals that other parts of the CPU can understand.

- Identifies which operation needs to be performed (arithmetic, logic, data transfer, etc.)

3. Controlling Data Flow
- Directs where data should move: between memory, ALU, registers, and I/O devices.

4. Generating Control Signals
- Sends Control signals to all hardware parts to tell them when and how to operate.

5. Maintaining Order of Execution
- Ensures instructions execute in the right sequence according to the program flow.

6. Handling Interrupts
- Detects when an interrupt occurs (such as an I/O request or error).

- Pauses the current task, handles the interrupt, then resumes the program.

- **Working Example - "2+3"**

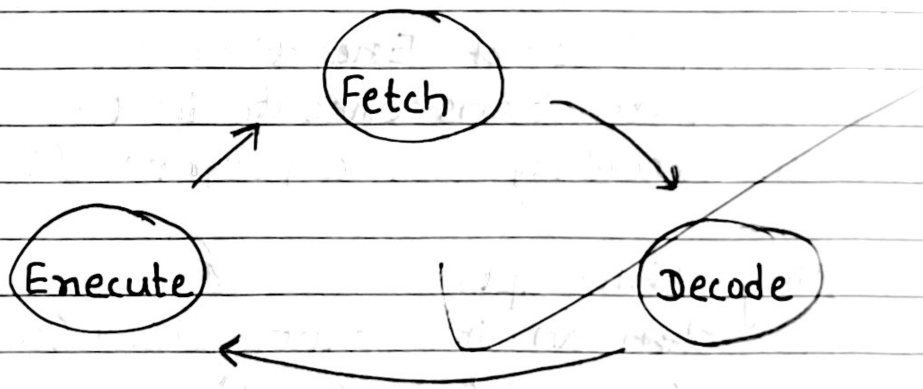
1. Fetch: CU Fetches instruction ADD 2,3 from memory.

2. Decode: CU interprets it as "send 2 and 3 to ALU for addition."

3. Control: CU sends signals to:
   - Load 2 into Register A
   - load 3 into Register B
   - Tell ALU to perform addition

4. Execute: ALU adds numbers and stores result in a register.

5. Next step: CU moves to the next instruction.



b) Arithmetic Logic Unit (ALU)

↳ The Arithmetic Logic Unit (ALU) is the part of CPU responsible for performing mathematical calculations and logical comparisons.

↳ It is the computational engine of the ~~control unit (CU), which~~ CPU - where all actual number crunching and decision-making happens.

↳ It works under the control of control Unit (CU), which tells it what operations to perform and on which data.

- Performs arithmetic operations (addition, subtraction) etc.

- Performs logical operations (AND, OR, NOT, comparisons).

↳ Registers

↳ Very small, high-speed storage locations inside CPU.

↳ Temporarily hold data, instructions, or addresses during processing.

3. <u>Output Unit</u>

↳ The Output Unit is the Part of the computer system that converts processed data from the CPU into a human-understandable form or a usable form for other systems.

↳ It works as the final communication bridge between the computer and the user (or another device).

• Main Functions

1. Receives Data from CPU
↳ Takes processed data (in binary form) from the CPU or memory.

2. Convert to Human-Readable Form
↳ Transforms binary data into text, graphics, audio, or physical form.

↳ Uses Digital-to-Analog Converters (DAC) for analog outputs like sound.

3. Display or Deliver Output
↳ Presents results to the user visually, audibly or physically.

4. Store Output
↳ Sends processed information to storage devices or other systems for later use.

• Types of Output

1. Soft Copy Output (Temporary, intangible)
↳ Seen or heard but not physically stored on paper.

↳ e.g ÷ Monitor / screen
÷ Projector display
÷ Audio output from Speakers.

2. Hard Copy Output (Permanent, Physical)
↳ Tangible and can be stored physically.

↳ e.g. : Printed documents
: Plotted ~~desig~~ diagrams
: 3D-Printed objects.

• Examples of Output devices
↳ Visual output
- Monitor / Display Screen - LCD, LED, OLED
- Projector for large scale visual display
- VR-Headset - Immersive visual experience.

↳ Printed Output
- Printer - Inkjet, laser, Dot matrix
- Plotter - For large technical drawings, blueprints.
- 3D printer - for physical models & prototypes.

↳ Audio Output
- Speakers - General sound output
- Headphones / Earbuds - Personal audio Outputs.

	First generation	Second generation	Third generation	Fourth generation	Fifth generation
Time	• (1946-1959)	• (1959 - 1965)	• (1965-1971)	• (1971 - 1980s)	• (1990s - Present)
Main Component	• Vacuum tube	• Transistor	• Integrated Circuit	• Microprocessor	• Bio chip
Primary Storage	• Magnetic drums	• Magnetic core memory	• Magnetic core technology • Semiconductor memory	• Semiconductor RAM [SRAM & DRAM]	• Advanced SRAM & DRAM, flash memory
Language	• Machine level language	• Assembly language	• High level language	• High level programming language	• Natural language processing
Input device	• Punched cards and paper tape	• Punched card, magnetic tape, keyboard	• Punched card, keyboard	• keyboard, mouse.	• touchscreen, Sensor, Cameras
Output device	• Printers & paper tape readers	• Printers and plotters.	• CRT monitors, Printer	• Color monitors, Plotters, printer	• LCD/LED display, 3D printers
features	• Larger in size, High cost, bulky, consumed more power	• Smaller in size, lesser power consumption, lower cost	• Smaller, better performance, comparatively lesser cost.	• Less power consumption, high performance, lower cost, very compact	• Faster and efficient computing, portable, affordable.
examples	• ENIAC, EDVAC, EDSAC, UNIVAC	• IBM 1401, IBM 1620, UNIVAC-II	• IBM 370, PDP-8, CDC 7600, NCR 395	• IBM PC, Cray-I, Intel PC	• Desktop, Laptop, Apple siri, etc.
Processing Speed	• Millisecond	• Microsecond	• Nanosecond	• ~~At~~ Picosecond	• More than Femto-second
Reliability and accuracy	• Not fully	• More than 1G	• Fully than Previous	• Fully than prévious	—

- Types of Computer on the Basis of size

1. Microcomputer

> Definition: A small, single-user computer that uses a microprocessor as its central processing unit (CPU).

> User Capacity: Designed for one user at a time.

> Purpose: Intended for personal or small-scale computing tasks.

> Examples: Desktop PCs, laptops, tablets, smartphones.

> Key Features:
- Low to moderate processing power compared to larger systems.
- Affordable and often portable.
- Runs general-purpose software like word processors, browsers, and media players.

> Uses:
- Word processing and document creation.
- Internet browsing and communication.
- Gaming and multimedia playback.

- office work and basic data management.

a) <u>Portable Microcomputers</u>

↳ Portable microcomputers are self-contained computing devices that are designed for easy transportation. They have an integrated display, keyboard or input interface, storage, and battery, allowing them to function without being plugged into a power source for a certain period.

↳ <u>Types and Examples</u>

- Laptops / Notebooks
  > Foldable computers with built-in display, keyboard, and touchpad (e.g. Dell XPS, MacBook Pro, HP Pavilion).

- Netbooks
  > Smaller and lighter than laptops, designed mainly for internet browsing and basic computing.

- Tablets
  > Flat, Touchscreen devices with virtual keyboards (e.g - Apple ipad, Samsung Galaxy Tab)

- Smartphones
  > Handheld devices combining mobile communication and computing power

(e.g. iphone, samsung Galaxy).

- PDAs (Personal Digital Assistants)
  > Older handheld Devices for personal scheduling and organization. (mostly obsolete).

- Palmtop
  > A palmtop is a compact computing device with a small screen, miniature keyboard or touch input, and built-in storage processing unit, and battery. It is designed for basic computing tasks that can be done on the go.

b. Non-Portable Microcomputers

↳ Non-Portable microcomputers are stationary computing systems that require external peripherals like monitors, keyboards, and mice. They do not have an integrated battery and are intended for use in one fixed location.

↳ Types and examples
- Desktop Computers
  > The most common non-portable microcomputers with separate monitor, CPU case, and peripherals (e.g. Dell Optiplex, HP Envy Desktop).

- Workstations
  > High performance desktops for tasks like 3D modeling, engineering simulations, and data analysis (e.g. lenovo Thinkstation, Dell precision).

- Gaming PCs
  > Customized desktops with high-end GPUs, CPUs and cooling systems for demanding games.

2. <u>Minicomputers</u>

> <u>Definition</u>: A mid-sized, multi-user computer larger than a microcomputer but smaller than a mainframe.

> <u>User Capacity</u>: Supports dozens to hundreds of users simultaneously.

> <u>Purpose</u>: Serves departmental or organizational needs.

> <u>Examples</u>: DEC PDP-11, VAX series (historical); modern equivalents including mid-range servers.

> <u>Key Features</u>:
  - Multi-user and Multi-tasking capabilities.

  - Higher processing power than microcomputers.

- Can store and process larger databases.

- Often used with specialized software for scientific or industrial purposes.

> Uses:
- Process control in factories.

- Database management for business or research.

- Business transaction processing for medium-sized organizations.

3. Mainframe Computers

> Definition: A large-scale, high-performance computer design for centralized data processing and supporting thousands of users at the same time.

> User Capacity: Capable of supporting thousands of simultaneous connections.

> Purpose: Manages mission-critical applications for large organizations.

> **Examples:** IBM Z-series, Unisys ClearPath.

> **Key Features:**
- Extremly reliable with minimal downtime.
- Handles massive input/output operations efficiently.
- Supports virtualization, allowing multiple systems to run at once.
- Designed for data security and high availability.

> **Uses:**
- Banking transactions and account management.
- Airline reservation systems.
- Government census data processing
- Large-scale enterprise resource planning (ERP).

A terminal in mainframe computing is a hardware device (or software emulation) that sends user commands to the mainframe and display the mainframe's output.

Types of Terminals

1. Dumb Terminals

> **Description**: Simple terminals with only a keyboard for input and a screen for output.

> **Processing Power**: No CPU or local storage; completely dependent on the mainframe.

> **Example**: IBM 3270 Terminal.

> **Uses**: Banking transactions, airline bookings, and centralized data entry.

2. Intelligent Terminals

> **Description**: Terminals with a small processor and limited memory for minor tasks like local editing or data buffering.

> **Processing Power**: Can handle small operations before sending data to mainframe.

> **Example**: DEC VT220 terminal.

> **Uses**: Reducing load on the mainframe for repetitive or preliminary processing.

4 Super Computers

> **Definition**: The fastest and most powerful computers in the world built for extremely complex, high-speed calculations.

> **User Capacity**: Often dedicated to one large-scale project at a time.

> **Purpose**: Executes scientific, engineering, and research simulations requiring enormous computing power.

> **Examples**: Fugaku (Japan), Frontier (USA), Summit (USA).

> **Key Features**:
- Capable of trillions to quadrillions of ~~comp~~ calculations per second (measured in FLOPS).

- Uses massive parallel processing with thousands of processors.

- Requires specialized cooling systems and large facilities.

- Optimized for complex ~~to~~ algorithms and simulations.

> Uses

- Weather forecasting and climate modeling.
- Nuclear simulations for safety and defense.
- Genome mapping and medical research.
- Space exploration and astrophysical simulations.

In supercomputers, FLOPS stands for Floating Point Operations per second - It is the standard unit used to measure how fast a supercomputer can perform calculations involving real numbers (floating point numbers).

05/05

• Assignment - 2

Write down the 7 difference between analog and digital signals.

	Analog Signal	Digital Signal
①	Analog signal is continuous in nature and changes smoothly over time.	Digital signal is discrete in nature and changes in steps or intervals.
②	It can take infinite values within a given range	It can take only finite values, usually represented

		as binary 0 and 1.
3	The waveform of an analog signal is represented by sine waves.	The waveform of a digital signal is represented by square waves.
4	It is more accurate in representing natural signals but is difficult to process.	It is less accurate due to quantization but easier to process with computer
5	Analog signals are more prone to noise and distortion during transmission.	Digital signals are less affected by noise and are more reliable for transfer.
6	Storing and processing analog signal is difficult and requires more equipment.	Storing compressing, and processing digital signals is easy and efficient.
7	Examples: Human voice, radio signals, thermometer readings.	Examples: Computer data, CDs, DVDs, digital telephones.

Mobile Computing : An In-Depth Overview

↪ Mobile computing refers to the human-computer interaction where computing devices are portable and can be used while in motion, enabling the transmission of data, voice, and video without fixed physical connections.

It encompasses a combination of mobile hardware (like smartphones and tablets), software (operating systems and applications), and communication technologies (wireless networks).

This allows users to access computation, information, and services anytime, anywhere, integrating elements like ~~prop~~ portability, connectivity, interactivity, and individuality to meet user needs in dynamic environments.

> Applications:-

- Communication and Social Media : Real-time messaging, video calls (e.g. whatsApp, zoom) and platforms like Instagram and twitter for global connection.

① ~~Communication and Social media:~~
~~-Real-time messaging, video calls~~

- Business and Productivity: Remote work tools (email, cloud storage like Google Drive), project management apps (Trello) and e-commerce (Amazon, mobile payments via Apple Pay).

- Healthcare: Telemedicine, remote patient monitoring, fitness tracking (e.g. Fitbit), and apps for electronic health records, enabling quick access to medical data.

- Education: E-learning platforms (Coursera, Khan Academy), virtual classrooms, and interactive for remote assessments and resource sharing)

- Entertainment and Multimedia:- Streaming services (Netflix, Spotify), gaming (PUBG Mobile)

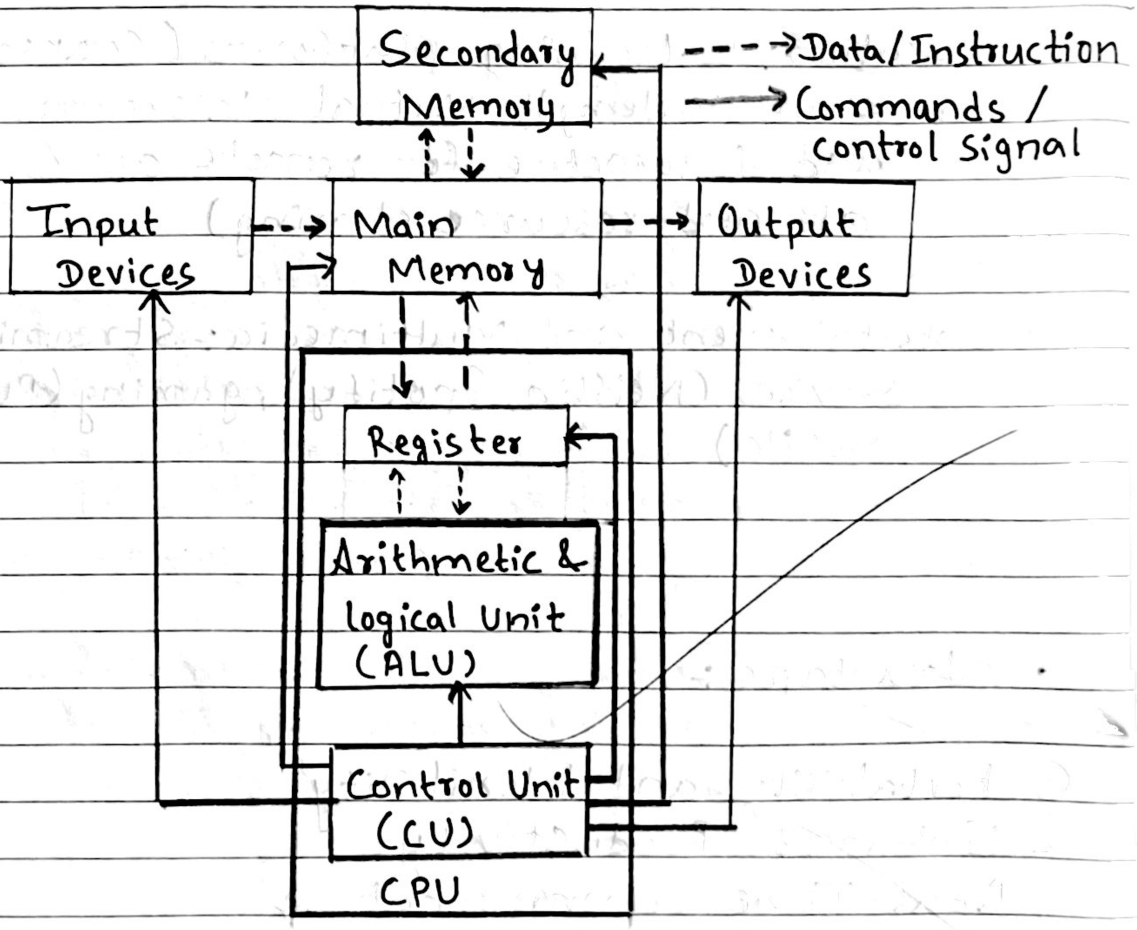
• Advantages :-

1. Portability and Flexibility
2. Enhanced Productivity
3. Real-Time Communication
4. Location-Based Services
5. Entertainment and Convenience
6. Economic benefits

- Disadvantages and challenges

1. Battery Life and Power Consumption
2. Security and privacy Risks
3. Connectivity issue
4. Device limitation
5. Health and Social Concerns

- Block diagram of a Computer System



Computer System

↳ An integrated set of hardware and software components that work together to process, store, and ~~manag~~ manage data to perform specific tasks.

Components of a Computer System

1. Input Unit ✓
2. Central Processing Unit ✓
3. Output Unit ✓
4. Memory Unit

↳

- Primary Memory

- Primary Memory, also called main memory, is the computer's fast, directly accessible storage used to hold data and instructions currently being processed by the CPU. It is volatile, meaning data is lost when the power is turned off.

Types of Primary Memory

a) RAM (Random Access Memory)

> Purpose - It stores data and instructions that the CPU needs in real time for running applications and processes.

> Characteristics:
- Volatile: Loses data when power is off.
- Fast read / write speeds.